YUXIN ZHANG

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PROFESSIONAL EXPERIENCE

Scam AI | San Francisco, CA

Founding Software Engineer | July 2026 – October 2026

- Owned a production **AI detection platform** as microservices (a Node.js/TypeScript API gateway, a FastAPI detection router, and GPU/CPU model servers), with multi-model routing, request batching, and async jobs via PostgreSQL and BullMQ.
- Deployed model-serving **infrastructure as code** with Terraform on GCP, provisioning autoscaling pools for 4.9× higher throughput.
- Built an in-house **model registry** and automated release pipeline for 4 production ML models, with versioned artifacts, CI-gated evaluations, and one-step rollback, enabling same-day model releases.
- Implemented an **observability** pipeline with PostgreSQL and BigQuery, centralizing API logs and performance metrics for troubleshooting.
- Engineered **distributed system** reliability with idempotent requests, a retry-aware error contract, Redis-backed token-bucket rate limiting, backpressure, and fail-closed handling; diagnosed and eliminated burst-traffic failures (32% → 0%) and load-tested at 600 requests/min.

Full-Stack Development Intern | May 2025 – August 2025; February 2026 – May 2026

- Optimized PyTorch **inference serving** through performance profiling, CUDA Graphs, mixed precision, and asynchronous CPU-GPU execution, reducing P95 inference latency by 15% and improving throughput by 20%.
- Developed containerized ML inference services on AWS (EC2, S3) using Python, FastAPI, and Docker, exposing standardized REST APIs.
- Shipped TypeScript/Python SDKs, CLI tooling, and MCP servers (stdio, Streamable HTTP), standardizing API-key authentication and secure access for third-party developers and AI agents.
- Engineered CI/CD pipelines with GitHub Actions and AI agents for **release validation**, reducing production deployment time by 47%.

University of Massachusetts Amherst | Amherst, MA

Research Assistant, BioNLP Lab | January 2025 – August 2025

- Built and evaluated GraphRAG pipelines for biomedical QA, combining LLM-guided retrieval and multi-hop reasoning over 1.5M+ knowledge-graph triples.
- Co-authored RiTeK, published in *Findings of ACL 2026*.

Huawei | Beijing, China

Software Engineer II, 2012 Lab | March 2022 – June 2024

- Architected an offline C++/Python **analysis pipeline** processing 10TB+ in parallel, reducing manual labor 40% and pipeline errors 65%.
- Created a C++ **profiling tool** for thread-level execution tracing to support performance analysis for next-generation chips.
- Co-developed a thread characterization method to guide **CPU scheduling** and dynamic frequency scaling, reducing battery use by 6%.
- Optimized CPU **cache allocation** in the Linux kernel, reducing application startup latency by 6% for 1.5M+ users.

Software Engineer, ICT Service & Software | October 2020 – March 2022

- Optimized ACL processing by simplifying execution paths, eliminating **redundant copies and allocations**, and reducing rule lookups via direct pointer access, cutting latency by 6% and improving throughput by 4%; debugged memory errors using AddressSanitizer (ASan).
- Implemented ACL and **traffic classification** features in the QoS module using C for enterprise routers serving Fortune 500 clients.
- Refactored the redirect module for modularity and maintainability, achieving 75% GoogleTest coverage and conducting code reviews.
- Integrated Jenkins CI/CD pipelines with automated unit and integration tests, reducing integration failures by 30%.

EDUCATION

University of Massachusetts Amherst | Amherst, MA

Master of Science in Computer Science | GPA: 4.0/4.0 | August 2024 – May 2026

University of Pittsburgh | Pittsburgh, PA

Bachelor of Science in Computer Science and Psychology | August 2016 – August 2020

TECHNICAL SKILLS

Languages: Python, TypeScript, C++, C, Java, SQL, JavaScript, HTML/CSS, Shell

Backend & Data: FastAPI, Node.js, Express.js, REST APIs, Microservices, PostgreSQL, Redis, MySQL, BigQuery

Cloud & DevOps: GCP (Cloud Run, Compute Engine), AWS (EC2, S3, Lambda, ECS), Docker, Terraform, CI/CD, Jenkins, Nginx

AI/ML Infrastructure: PyTorch, CUDA, CUDA Graphs, MCP, RAG